Experiment ledger and glossary

Primary comparison incomplete; no claim about an oscillator advantage is supported.

All networks generate unconditional images in pixel space. They share independent conditional flow matching, AdamW, continuous time conditioning, 4×4 patches, and an EMA sampling checkpoint. MNIST is development data; CIFAR-10 is the final comparison domain. There is no curriculum or augmentation. Classification accuracy, text fill rate, and task-completion accuracy are not applicable; run completion is reported separately from image quality.

Network	Feature computation	Training objective	Role
onn	Coupled 4D unit-vector rotations; four recurrent blocks	Velocity MSE	Proposed generator
dit	Four time-conditioned attention/MLP blocks	Velocity MSE	Parameter-matched transformer
conv	Convolution + SiLU + GroupNorm recurrence	Velocity MSE	Conventional recurrent control
no_coupling	Oscillators with coupling convolution removed	Velocity MSE	Trained removal; lower capacity
no_frequency	Oscillators with natural-frequency term removed	Velocity MSE	Trained removal; slightly lower capacity

Training outcomes

Run	Status	Parameters	Updates	Train seconds	Images/s	Validation MSE
checks/cifar10-metric-smoke	untrained	1.29771e+06	0	0	—	—
checks/mnist-smoke	step_limit	1.28949e+06	100	33.9622	376.889	2.00869
study/pilot/mnist-conv-s101	completed	1.28846e+06	5245	1200.05	559.445	0.215197
study/pilot/mnist-dit-s101	completed	1.29064e+06	6778	1200.04	722.965	0.203531
study/pilot/mnist-onn-s101	completed	1.28949e+06	3736	1200.25	398.425	0.231511

Distribution and cost measurements

Evaluation	Generated/real N	NFE	KID mean	KID subset SD	FID	Pixel SWD	ms/image
checks/cifar10-metric-smoke/evaluations/validation	128/5000	32	0.609741	0.0059603	487.666	—	21.3627
checks/mnist-smoke/validation-check	128/5000	32	—	—	—	1.10101	20.9486
study/pilot/mnist-conv-s101/evaluations/validation	512/5000	32	—	—	—	0.063441	13.336
study/pilot/mnist-dit-s101/evaluations/validation	512/5000	32	—	—	—	0.0617331	9.42476
study/pilot/mnist-onn-s101/evaluations/validation	512/5000	32	—	—	—	0.0714237	27.8529

Completeness

Recorded diagnostics: 1. Recorded failed attempts: 1. Completed primary pairs: 0/6. Missing values mean unmeasured, not zero. Individual records and PDFs are in `experiments/`; raw records remain in the run directories and published artifact manifests.

Glossary

Term	Meaning
Flow matching	Learn a velocity that transports Gaussian noise toward the image distribution.
Oscillator	A hidden vector whose direction changes under natural rotation and learned interactions.
Coupling	A learned convolution that lets hidden oscillators influence one another.
Flow time / internal time	Noise-to-image position / recurrent computation inside one velocity prediction.
NFE	Number of velocity-network evaluations used to generate each image.
KID	Unbiased squared kernel discrepancy between fixed Inception features; smaller is better and estimates may be negative.
KID subset SD	Variation across random metric subsets of one fixed pair of image populations, not across trained models.
FID	Gaussian feature-distribution distance; finite-sample biased, so sample count must accompany it.
Pixel SWD	Wasserstein distance averaged over random pixel-space projections; a MNIST diagnostic, not semantic accuracy.
Seed pair	Two independently initialized architectures using paired data/noise/time stream prefixes.
Confidence interval	A model-based repeated-study procedure; here an approximate t interval across six seed differences.
Sign-flip test	Enumerate all signs of paired differences; exact only under the specified exchangeability/symmetry null.
EMA	Exponential moving average of trainable weights, used for all reported samples.
Ablation / intervention	Retrain with a mechanism removed / perturb a trained model during inference.
Power	Probability of detecting an assumed effect under a planning model; six pairs have about 24% power at effect 0.8.